Aarhus University

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Multiplayer Analysis

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1. Multiplayer Analysis

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This Analysis was performed with the purpose of investigating different types of online multiplayer topologies. The types researched were **Local Area Network (LAN)**, **Peer-to-Peer (P2P)** and **Dedicated Servers**, with **Local multiplayer (Split Screen)** being tested in a POC.

1.1. Components

Several components are part of multiplayer implementation

Transport layer

- Lowest levet
- Responsible for transporting data between players

Runtime

- Game client
- Game server

Backend

- Matchmaking
- Databases

Fleet

- Servers for hosting

1.2. Topologies

1.2.1. Split Screen

This topology is tested in the split screen section, and only uses the Runtime component.

Pros:

- 0 Latency
- 0 Cost (Servers etc.)

Cons:

- Max 4 players per game (More introduces cluttering)
- Have to be on the same screen (Removes the communication aspect of the different perspectives)

Conclusion: Since the asymmetric perspective and communication is a big part of the game, split screen is not a viable solution for a standalone multiplayer experience.

1.2.2. LAN

Utilizes the transport layer for sending data between the players and the runtime for implementation.

Pros:

- Very low latency (Practically 0)
- 0 Cost (Servers etc.)
- Simple to implement
- Scales well

Cons:

- Limited to playing with people in the same location

Conclusion: If LAN is not too difficult to implement, it is a very viable solution for an MVP and as a separate game mode. It lets players connect to each other directly and play with very low latency, while persisting separate screens.

1.2.3. P2P

Uses the transport layer, the runtime and the backend.

There are two main types of P2P design.

Direct P2P — Each player is connected to every other player.

Pros:

- Cheap

Cons:

- Complex
- Does not scale well
- Security (IP’s exposed among players)

Conclusion: Direct P2P is a viable option, since the game will only have two players at a time. The increased complexity however, is something to note and must be taken into account. Security is not an issue in the MVP.

Player-hosted P2P (client/server) — One player runs both the client and server runtime, and acts as a host.

Pros:

- Cheap

Cons:

- Host advantage
- Host migration
- Host can cheat easily

Conclusion: Player-hosted P2P is currently the most fitting solution. It is cheap for our MVP, not too hard to implement, and host advantage/cheating is a non-issue, since it is a cooperative game, as opposed to a competitive one. Host migration should be non-frequent, since it is only two players.

1.2.4. Dedicated Servers

Uses all components

Pros:

- Flexibility in performance, scalability, global distribution etc.

Cons:

- Cost
- Maintenance
- Complexity

Conclusion: Dedicated servers are a very tempting solution. It fits all our criteria, and keeps latency low between players that play over large distances. However, the added complexity and cost, makes it an unviable solution in this project.